

ELEC4631 Lecture 13: State and Output Feedback Control for Reference Tracking

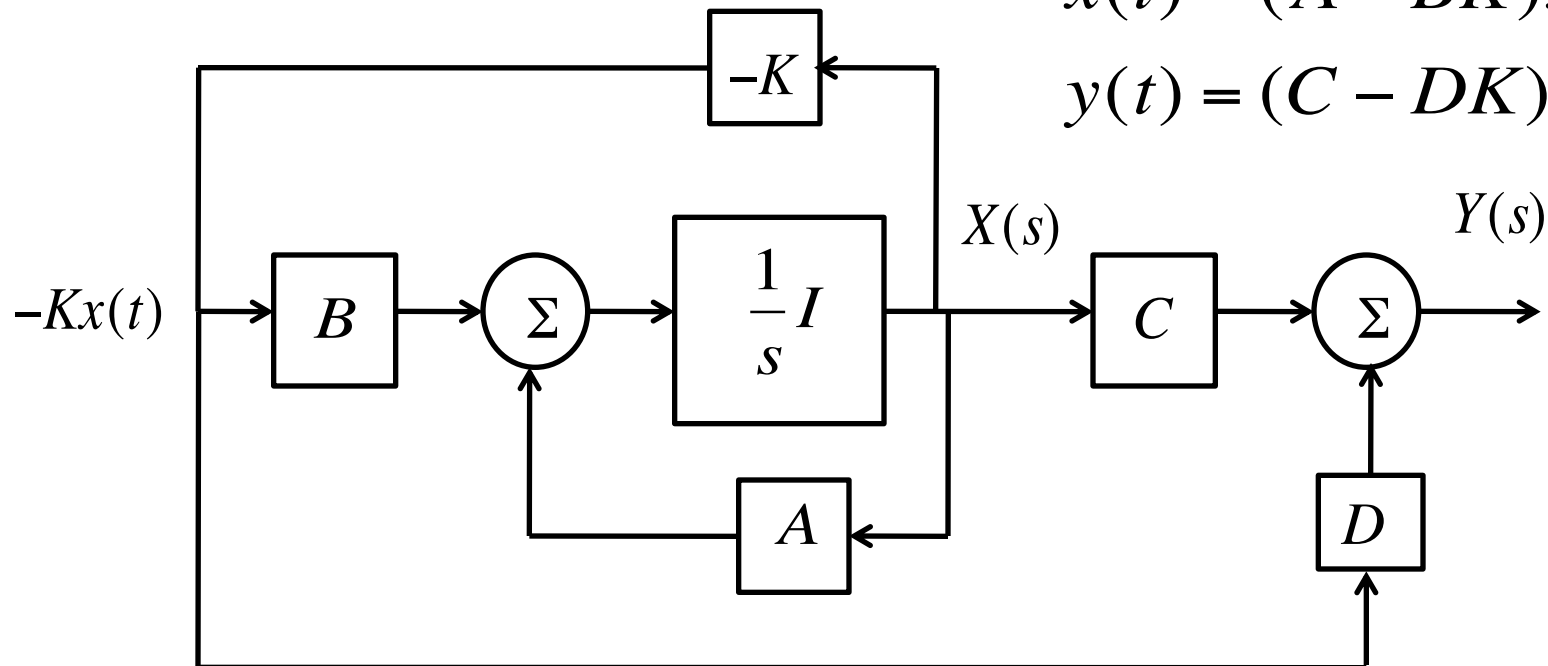
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State feedback

- State feedback primarily for stabilising system and modifying its transient dynamics.

$$\dot{x}(t) = (A - BK)x(t)$$

$$y(t) = (C - DK)x(t)$$

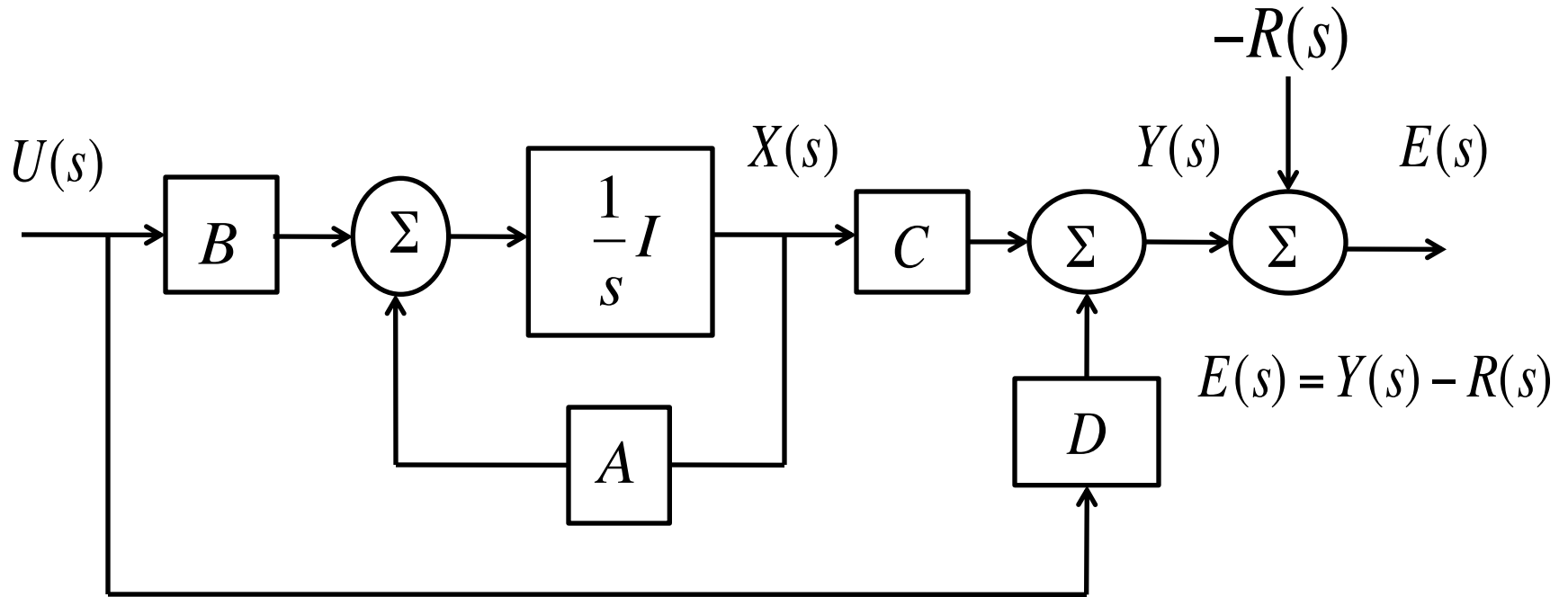


State feedback

- Control system first and foremost must ensure internal stability.
- Thus why emphasis on design of stabilising gain K (three approaches).
- Stabilising feedback controllers can be viewed as “resetting” controllers – resetting state to 0.

Reference tracking

- However, typically want controller to force plant output to follow a reference/command signal $r(t)$.

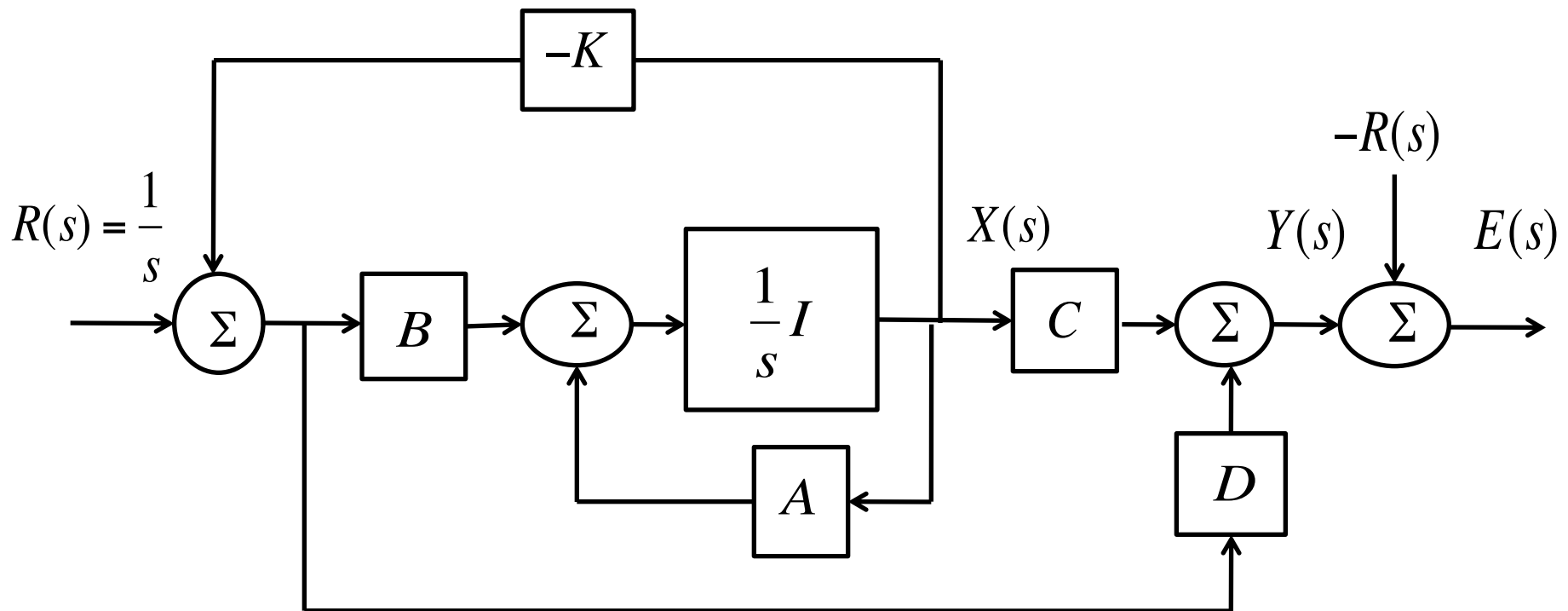


State feedback

- How to find $u(t)$ so $y(t)$ follows $r(t)$? This is *reference tracking problem*.
- Design methods for stabilising controllers
foundation for design of tracking controllers.
- First see how to track with state feedback for step reference signal $r(t)$.

Reference tracking

- Consider control signal of the form $u(t) = -Kx(t) + r(t)$, $r(t)$ is unit step input $r(t) = 1(t)$.



Reference tracking with state feedback

- So,

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ &= Ax(t) + B(-Kx(t) + r(t)) \\ &= (A - BK)x(t) + Br(t) \\ y(t) &= Cx(t) + D(-Kx(t) + r(t)) \\ &= (C - DK)x(t) + Dr(t)\end{aligned}$$

Reference tracking with state feedback

- If K chosen to be stabilising ($A-BK$ is stable matrix, all eigenvalues in LHP):

$$\dot{x}(t) \rightarrow 0$$

$$x(t) \rightarrow x_{ss}$$

$$y(t) \rightarrow y_{ss} = (C - DK)x_{ss} + D$$

as $t \rightarrow \infty$.

Reference tracking with state feedback

- So, this gives two steady-state equations

$$0 = (A - BK)x_{ss} + B$$

$$y_{ss} = (C - DK)x_{ss} + D$$

- In general, $y_{ss} \neq 1$.
- To guarantee $y_{ss} = 1$ need to modify input slightly.

Reference tracking with state feedback

- Suppose $u(t)$ settles to u_{ss} as $t \rightarrow \infty$. Consider control law

$$u(t) = u_{ss} - K(x(t) - x_{ss})$$

- Now, in steady-state:

$$\begin{aligned} 0 &= Ax_{ss} + Bu_{ss} \\ 1 &= Cx_{ss} + Du_{ss} \end{aligned} \quad \Rightarrow \quad \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x_{ss} \\ u_{ss} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Reference tracking with state feedback

- If

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

invertible, can solve for x_{ss} and u_{ss} :

$$\begin{bmatrix} x_{ss} \\ u_{ss} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Reference tracking with state feedback

- Thus with

$$\begin{aligned}u(t) &= u_{ss} - K(x(t) - x_{ss}) \\ &= (u_{ss} + Kx_{ss}) - Kx(t)\end{aligned}$$

we get $y_{ss} = 1$ as steady state output $\rightarrow$ output tracks unit step input.

- If reference is step with amplitude c ($r(t) = c1(t)$) then $y_{ss} = c$ because system is linear.

Reference tracking with state feedback

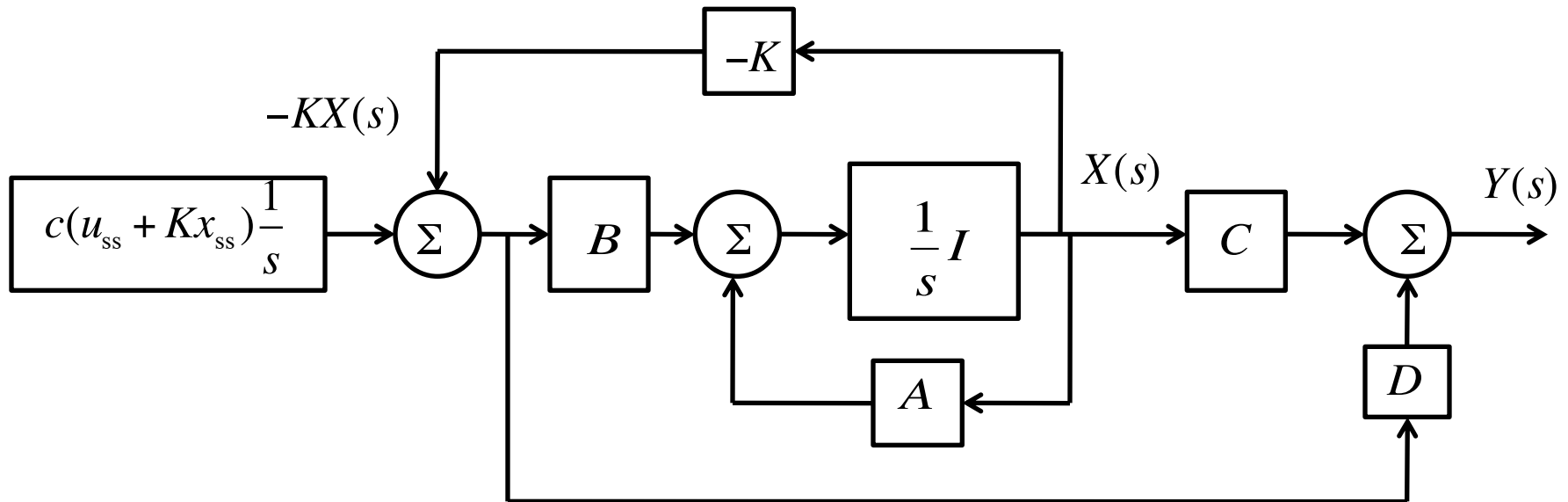
- If (A,B) controllable and

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

not invertible there is no solution $\rightarrow$ system cannot track step input.

Reference tracking with state feedback

- With configuration below, $y(t)$ tracks step input $r(t) = c1(t)$ in steady state for any step amplitude c .



Example

- Linearised model of simple pendulum with oscillation frequency $\omega_o = 1$ is

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -\omega_o^2 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)$$

- Control design goal: (a) both closed-loop poles are at $-2\omega_o$, and (b) output $y(t) = x_1(t)$ follows a step command.

Example

- Here gain is $K = [3 \ 4]$ by Ackermann's. To determine x_{ss} and u_{ss} solve:

$$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_{ss} \\ u_{ss} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(here x_{ss} is a 2 x 1 vector). Solving, we get

$$x_{ss} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad u_{ss} = 1$$

Reference tracking with output feedback

- So, with state feedback control law $u(t) = c(u_{ss} + Kx_{ss})1(t) - Kx(t)$ plant can track step input $r(t) = c1(t)$ for any amplitude c .
- Can this be done in typical situation of output feedback where only output $y(t) = Cx(t)$ accessible?

Reference tracking with output feedback

- Yes, it can be done.
- There are different approaches (e.g., Chapter 7 of text of Franklin, Powell and Emami-Naeni).
- We consider only one approach: *output feedback with a zero assignment estimator*.

Zeros of state-space systems

- Result from linear algebra: if M is an invertible matrix then

$$\det\left(\begin{bmatrix} M & N \\ P & Q \end{bmatrix}\right) = \det(M)\det(Q - PM^{-1}N)$$

$Q - PM^{-1}N$ is Schur complement of $\begin{bmatrix} M & N \\ P & Q \end{bmatrix}$ with respect to sub-matrix M .

Zeros of state-space systems

- Therefore, if $sI - A$ is invertible:

$$\det \begin{pmatrix} sI - A & -B \\ C & D \end{pmatrix} = \det(sI - A)(D + C(sI - A)^{-1}B)$$

$$H(s) = C(sI - A)^{-1}B + D = \frac{\det \begin{pmatrix} sI - A & -B \\ C & D \end{pmatrix}}{\det(sI - A)}$$

Polynomial for zeros

Transfer function from input to output

Characteristic polynomial

Zeros of state-space systems

- So, zeros of transfer function $H(s)$ are solutions of

$$\det\left(\begin{bmatrix} sI - A & -B \\ C & D \end{bmatrix}\right) = 0$$

Example: Zeros of state-space systems

- Consider state-space system

$$S = \left(\begin{bmatrix} -4 & 1 \\ 1 & -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 & 2 \end{bmatrix}, 1 \right)$$

- The zeros are roots of the polynomial $z_S(s)$

$$z_S(s) = \det \begin{pmatrix} sI - \begin{bmatrix} -4 & 1 \\ 1 & -2 \end{bmatrix} & -\begin{bmatrix} -3 \\ 3 \end{bmatrix} \\ \begin{bmatrix} 5 & 2 \end{bmatrix} & 1 \end{pmatrix} = s^2 - 3s + 10$$

Example: Zeros of state-space systems

- System transfer function $H_s(s)$ is therefore

$$H_s(s) = \frac{z_s(s)}{\det(sI - A)} = \frac{s^2 - 3s + 10}{s^2 + 6s + 7}$$

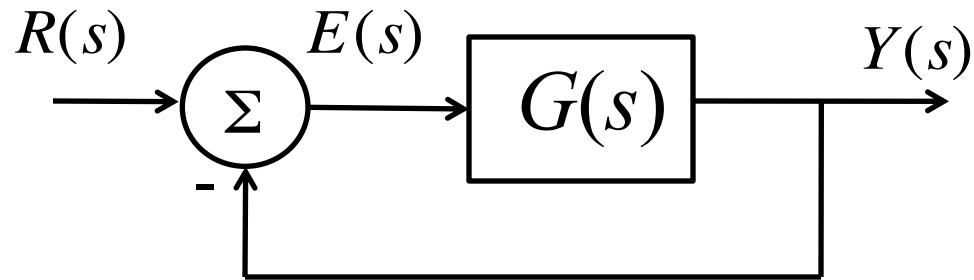
Type 1 system

- Recall some basic concepts from ELEC3114.
- System with transfer function $G(s)$ is Type 1 if it has one integrator

$$G(s) = \frac{1}{s} G_0(s)$$

$G_0(s)$ does not have 0 as pole or zero.

Type 1 system



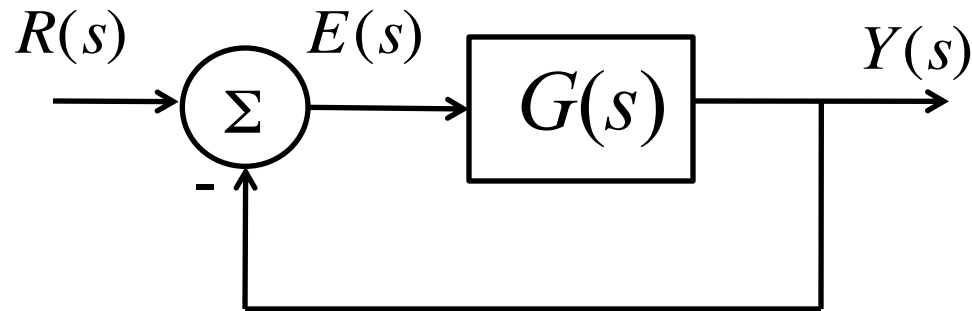
- If system Type 1, $R(s) = s^{-1}$ (step input), and closed-loop poles in LHP,

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) = 0$$

Final Value Theorem

- Type 1 system follows step input with no error.

Type 1 system

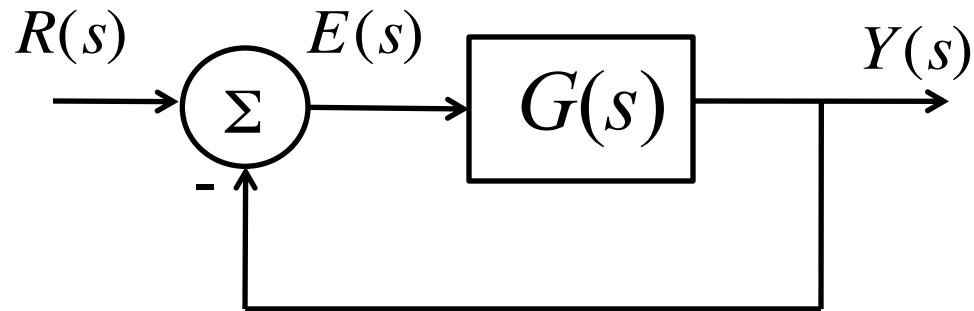


- If $R(s) = s^{-2}$ (ramp input),

$$0 < \left| \lim_{t \rightarrow \infty} e(t) \right| = \left| \lim_{s \rightarrow 0} sE(s) \right| < \infty$$

- Type 1 system follows ramp with finite non-zero steady-state error.

Type 1 system



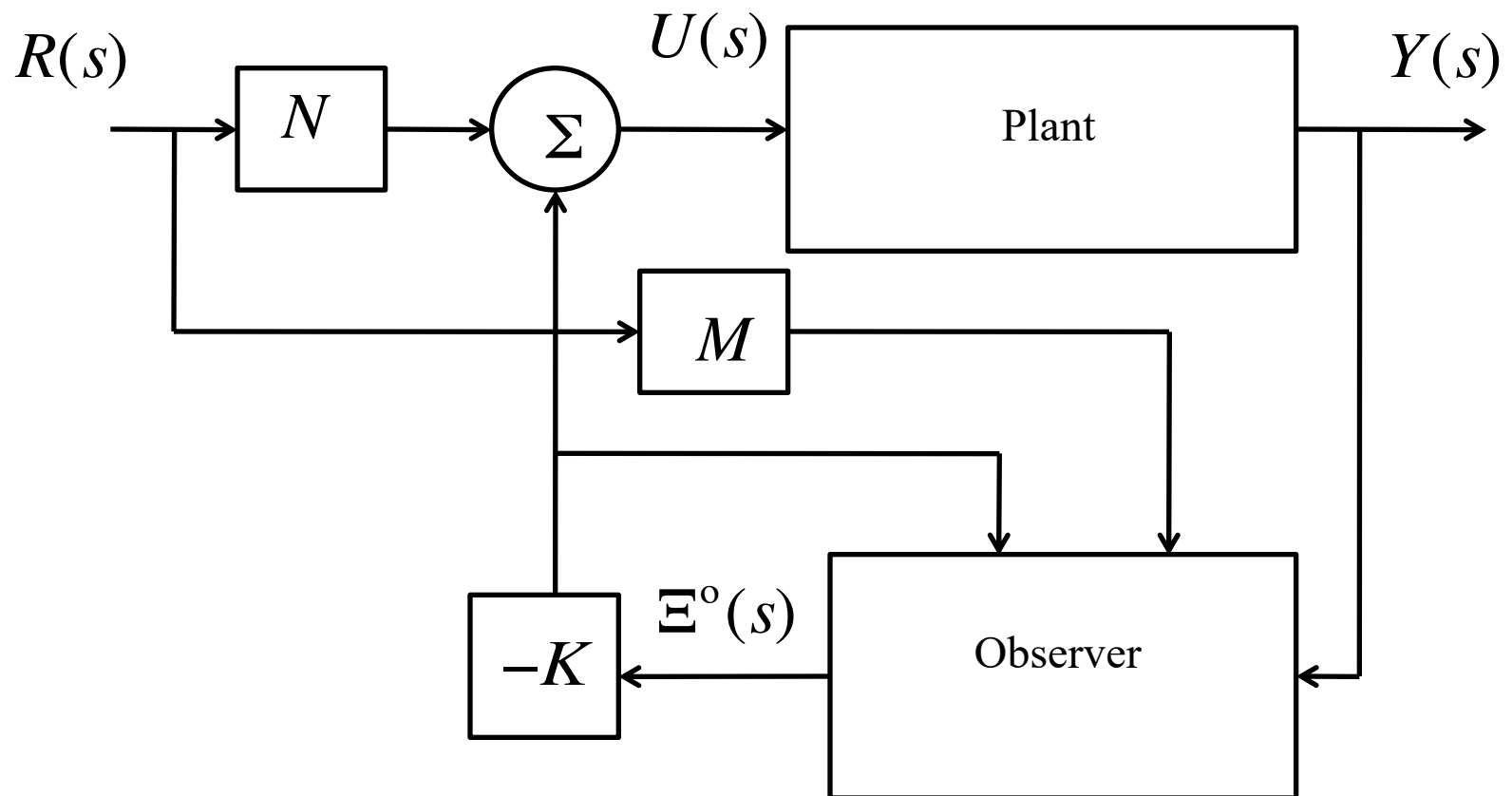
- Static error velocity constant K_v is

$$K_v = \lim_{s \rightarrow 0} sG(s) = G_0(0)$$

- Error of Type 1 for ramp $R(s) = s^{-2}$ is $1/K_v$.

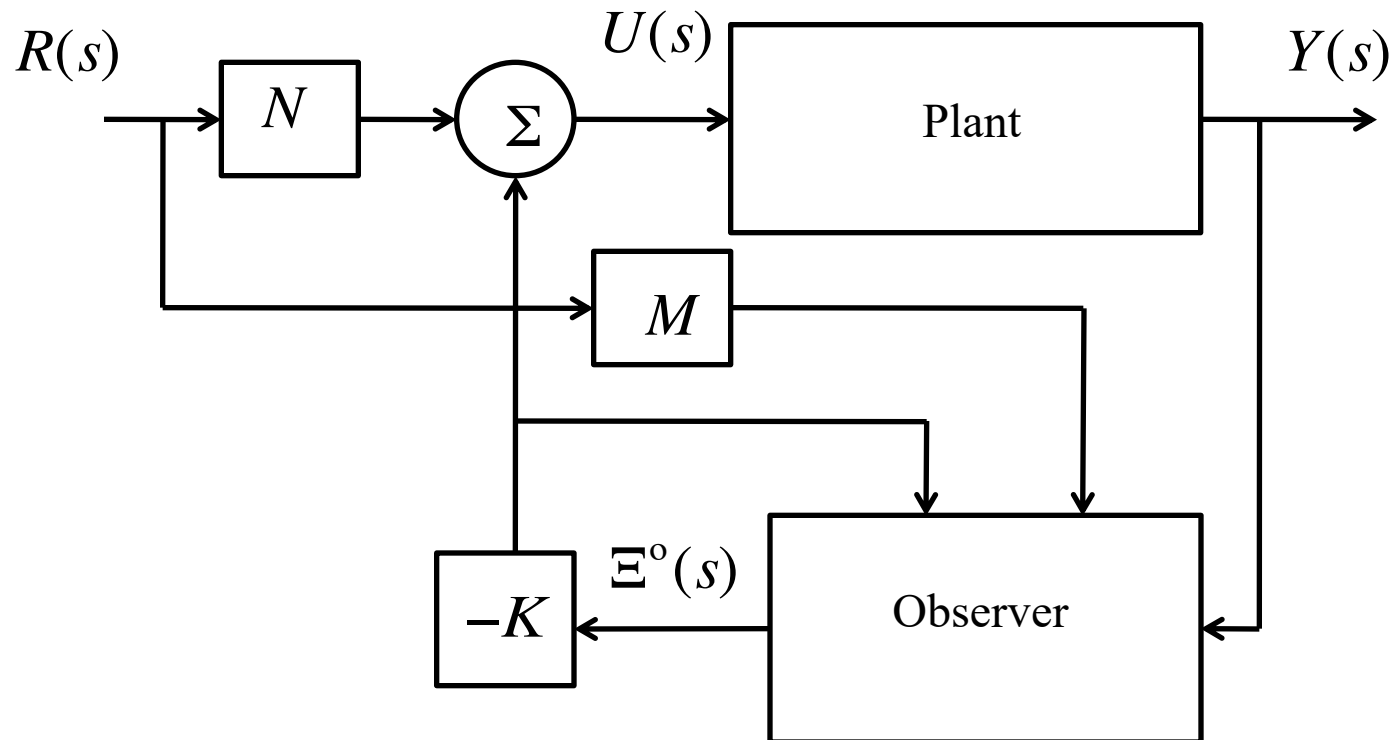
Output feedback with a zero-assignment estimator

- General structure:



Output feedback with a zero-assignment estimator

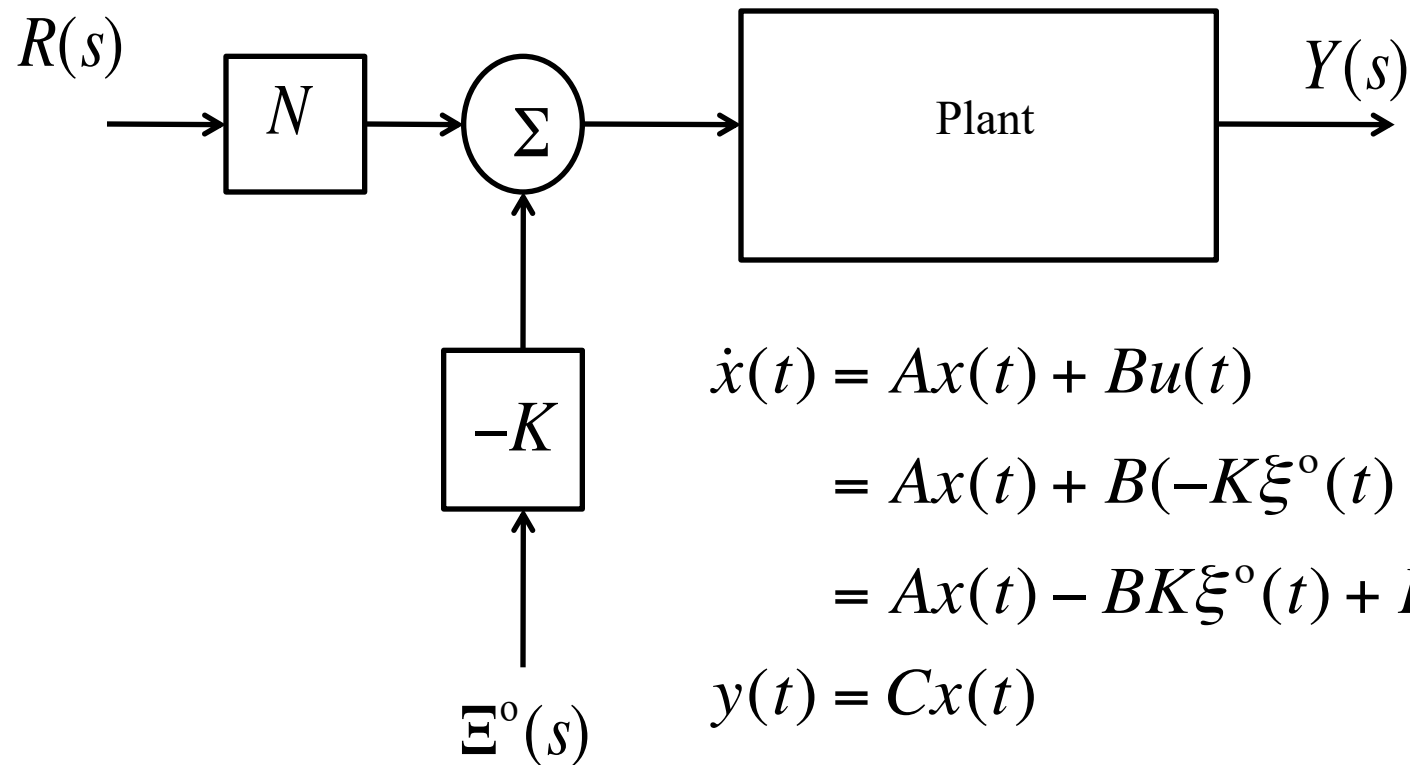
- N is non-zero scalar and M is vector to be chosen by control system designer.



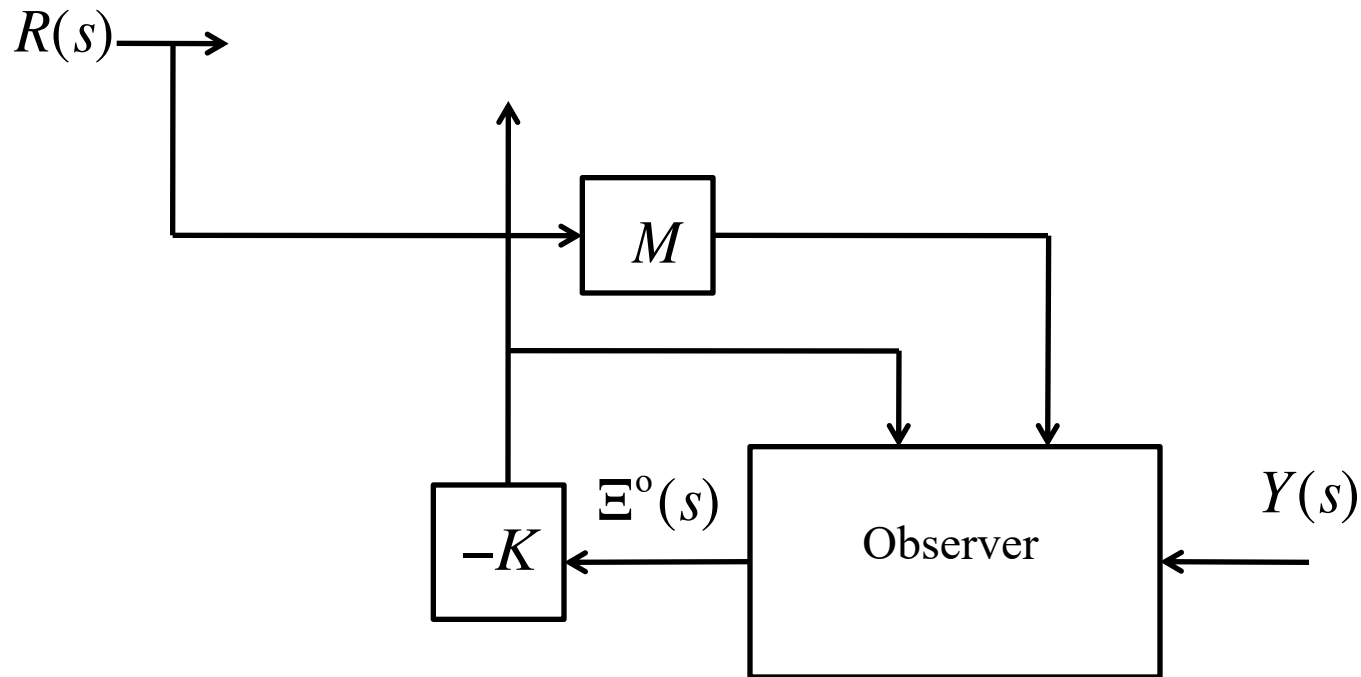
Design philosophy: output feedback with a zero-assignment estimator

- Design philosophy is as follows:
 - Set feedback gain K to place dominant closed-loop poles.
 - Set observer gain L and the location of some closed-loop zeros to achieve a desired K_v .
 - Set the ratio M/N to place part of the closed-loop zeros of the transfer function from $R(s)$ to $Y(s)$.
 - Set N to make closed-loop DC gain 1 so system follows step input.

Plant sub-dynamics



Observer sub-dynamics



$$\begin{aligned}\dot{\xi}^o(t) &= A\xi^o(t) - BK\xi^o(t) + L(y(t) - C\xi^o(t)) + Mr(t) \\ &= (A - BK - LC)\xi^o(t) + Ly(t) + Mr(t)\end{aligned}$$

Error dynamics

$$e(t) = x(t) - \xi^o(t)$$

$$\dot{e}(t) = \dot{x}(t) - \dot{\xi}^o(t)$$

$$\begin{aligned} &= Ax(t) - BK\xi^o(t) + BNr(t) - ((A - BK - LC)\xi^o(t) \\ &\quad + Ly(t) + Mr(t)) \end{aligned}$$

$$\begin{aligned} &= Ax(t) - BK\xi^o(t) + BNr(t) - ((A - BK - LC)\xi^o(t) \\ &\quad + LCx(t) + Mr(t)) \end{aligned}$$

$$= (A - LC)(x(t) - \xi^o(t)) + BNr(t) - Mr(t)$$

$$= (A - LC)e(t) + (BN - M)r(t)$$

Error dynamics

- So,

$$\begin{bmatrix} \dot{x}(t) \\ \dot{e}(t) \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x(t) \\ e(t) \end{bmatrix} + \begin{bmatrix} BN \\ BN - M \end{bmatrix} r(t)$$

Error dynamics

- Since

$$\begin{bmatrix} x(t) \\ e(t) \end{bmatrix} = \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} \begin{bmatrix} x(t) \\ \xi(t) \end{bmatrix} \Leftrightarrow \begin{bmatrix} x(t) \\ \xi(t) \end{bmatrix} = \begin{bmatrix} I & 0 \\ I & -I \end{bmatrix} \begin{bmatrix} x(t) \\ e(t) \end{bmatrix}$$

the equation

$$\begin{bmatrix} \dot{x}(t) \\ \dot{e}(t) \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x(t) \\ e(t) \end{bmatrix} + \begin{bmatrix} BN \\ BN - M \end{bmatrix} r(t)$$

implies

$$\lambda_{cl}(s) = \det(sI - A + BK) \det(sI - A + LC)$$

Error dynamics

- Identity

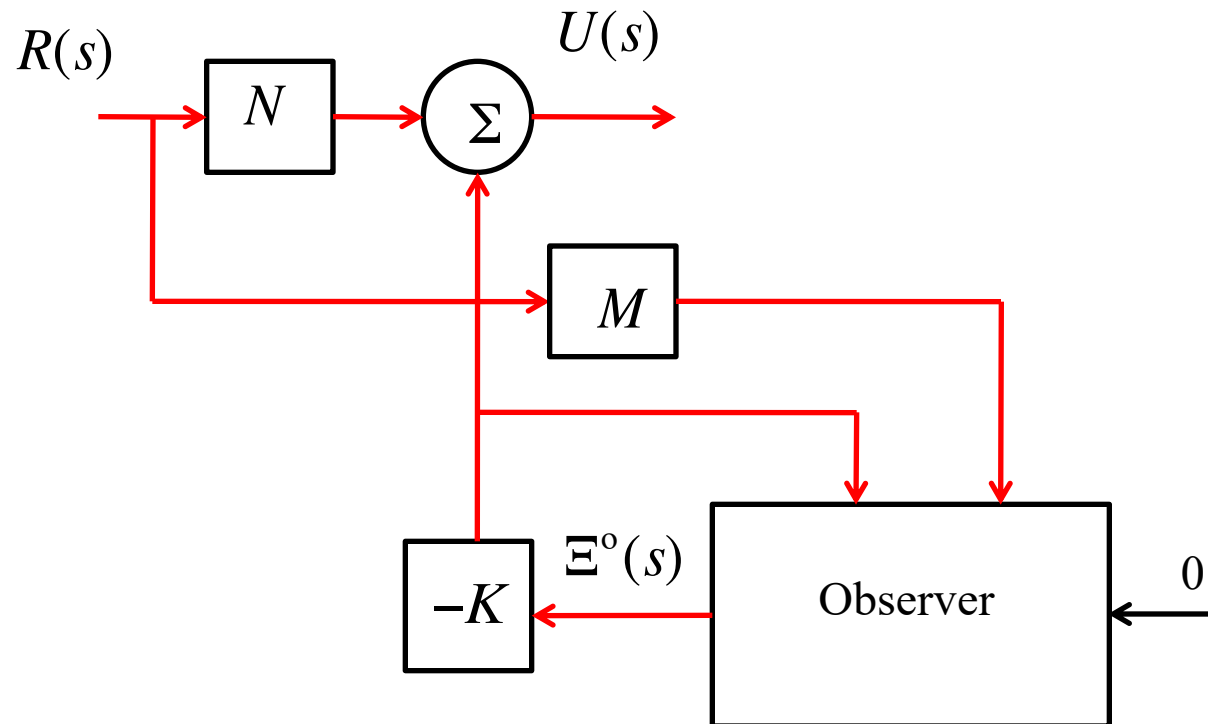
$$\lambda_{cl}(s) = \det(sI - A + BK) \det(sI - A + LC)$$

is expression of separation principle.

- That is, **separation principle (Lecture 10)** also holds for present output feedback control scheme.
- Feedback gain K and observer gain L for scheme can be designed *separately* and *independently*.

Transfer function from $R(s)$ to $U(s)$

- To find transfer function $T_{RU}(s)$ from $R(s)$ to $U(s)$, set $Y(s) = 0$.



Transfer function from $R(s)$ to $U(s)$

- Then

$$\Xi^o(s) = (sI - A + BK + LC)^{-1} MR(s)$$

$$U(s) = -K\Xi^o(s) + NR(s)$$

$$= (-K(sI - A + BK + LC)^{-1} M + N)R(s)$$

$$T_{RU}(s) = \frac{U(s)}{R(s)} = -K(sI - A + BK + LC)^{-1} M + N$$

Transfer function from $R(s)$ to $U(s)$

- The zeros of $T_{RU}(s)$ are solutions to

$$\det \begin{bmatrix} sI - A + BK + LC & -M \\ -K & N \end{bmatrix} = 0$$

- The left hand side is a polynomial in s .

Transfer function from $R(s)$ to $U(s)$

- Now, for non-zero N :

$$\det \begin{bmatrix} sI - A + BK + LC & -M \\ -K & N \end{bmatrix} = 0$$



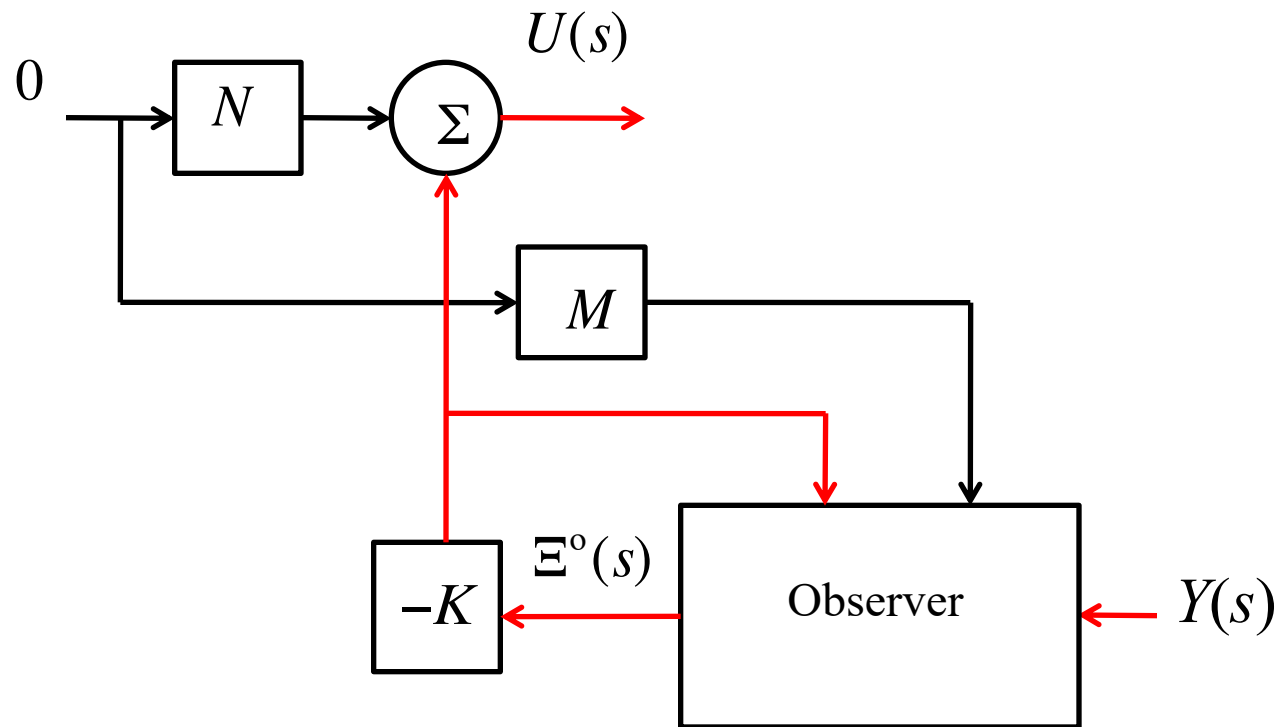
$$\det \begin{bmatrix} sI - A + BK + LC - \frac{M}{N}K & -M/N \\ 0 & 1 \end{bmatrix} = 0$$



$$\gamma(s) = \det \left(sI - A + BK + LC - \frac{M}{N}K \right)$$

Transfer function from $Y(s)$ to $U(s)$

- Similarly, to find transfer function $T_{YU}(s)$ from $Y(s)$ to $U(s)$, set $R(s) = 0$.



Transfer function from $Y(s)$ to $U(s)$

- Then

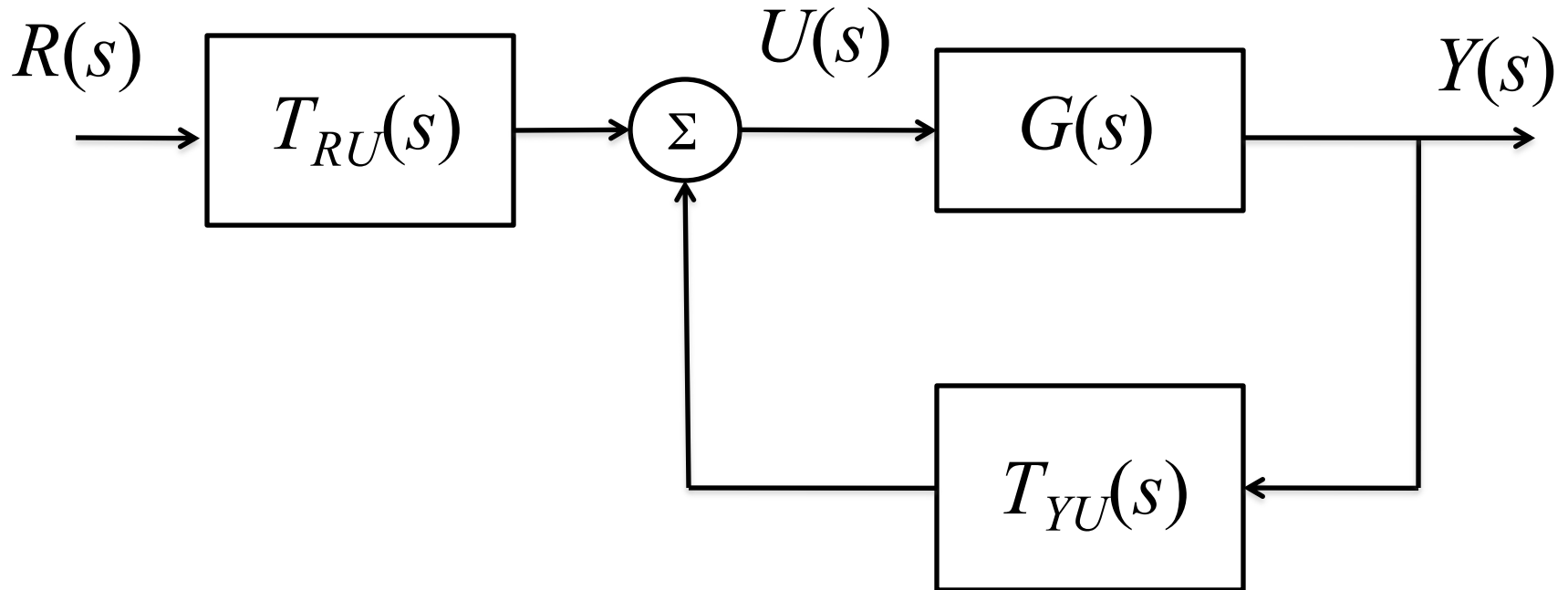
$$\Xi^o(s) = (sI - A + BK + LC)^{-1}LY(s)$$

$$U(s) = -K\Xi^o(s)$$

$$= -K(sI - A + BK + LC)^{-1}LY(s)$$

$$T_{YU}(s) = \frac{U(s)}{Y(s)} = -K(sI - A + BK + LC)^{-1}L$$

Equivalent closed-loop block diagram



$$H_{cl}(s) = \frac{T_{RU}(s)G(s)}{1 - G(s)T_{YU}(s)}$$

Zeros from $R(s)$ to $Y(s)$

- So, zeros of $H_{cl}(s) = Y(s)/U(s)$ are zeros of $T_{RU}(s)$ plus the zeros of $G(s)$.
- Thus, the closed-loop zeros are the roots of $\gamma(s)$ plus the zeros of $G(s)$.
- $\gamma(s)$ gives partial control of location of system zeros to complement zeros from $G(s)$.

Designing M and N

- Since zeros of $\gamma(s)$ are also zeros of $Y(s)/U(s)$ and

$$\gamma(s) = \det \left(sI - A + BK + LC - \frac{M}{N}K \right)$$

means zeros of $Y(s)/U(s)$ partially depend on M and N .

- Free parameters M and N designed to set part of zeros of transfer function from input to output.

Designing M and N

- Actually, $\gamma(s)$ does not depend on M and N individually but on quotient $\beta = M / N$.

$$\gamma(s) = \det(sI - A + BK + LC - \beta K)$$

- So, design of $\gamma(s)$ is to choose the right β .

Designing M and N

- Assume closed-loop system can track a step input (closed-loop DC gain is 1).
- If closed-loop poles of system are $\{p_k\}$ and closed-loop zeros are $\{z_k\}$ then K_v is (Truxal's Formula):

$$\frac{1}{K_v} = \sum \frac{1}{z_k} - \sum \frac{1}{p_k}$$

- Zeros of $\gamma(s)$ can be chosen to achieve desired K_v .

Truxal's Formula

- Any system with closed-loop SISO transfer function $H_{cl}(s)$ with DC gain 1 (tracks a step input) can be expressed as unity feedback loop with some Type 1 system $G(s)$ in the loop

$$H_{cl}(s) = \frac{G(s)}{1 + G(s)}$$

- Truxal's Formula says that by appropriately setting the locations of poles and zeros of $H_{cl}(s)$ one can set K_v to a desired value.

Designing M and N

- If $|z_k - p_k|$ small then this zero-pole pair has little effect on the system's dynamic response because they almost cancel each other.
- Yet, even if $|z_k - p_k|$ small $1/z_k - 1/p_k$ can be substantial and have non-negligible effect on K_v via Truxal's Formula.
- In choosing z_k and p_k , we can try to make these two effects work *simultaneously*.

Designing M and N

- Once $\beta = M/N$ chosen to get $\gamma(s)$, still have one free parameter.
- Take remaining free parameter to be N . (M is then determined by $M = \beta N$).
- Pick N such that closed-loop DC gain, $H_{cl}(0)$, is 1. This ensures $y(t)$ tracks unit step input $r(t)$.

Designing M and N

- How to determine N ? Combined equations for $x(t)$, $e(t)$ and $y(t)$ is (from earlier)

$$\begin{bmatrix} x(t) \\ e(t) \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x(t) \\ e(t) \end{bmatrix} + \begin{bmatrix} B \\ B - \beta \end{bmatrix} Nr(t)$$

$$y(t) = [C \quad 0] \begin{bmatrix} x(t) \\ e(t) \end{bmatrix}$$

Designing M and N

- For closed-loop DC gain of 1:

$$H_{cl}(0) = -[C \quad 0] \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix}^{-1} \begin{bmatrix} B \\ B - \beta \end{bmatrix} N = 1$$

- So,

$$N = -\frac{1}{C(A - BK)^{-1}B(1 - K(A - LC)^{-1}(B - \beta))}$$

Closed-loop transfer function

- The closed-loop transfer function $H_{cl}(s)$ has the form:

$$H_{cl}(s) = \frac{K_{cl} z_G(s) \gamma(s)}{\det(sI - A + BK) \det(sI - A + LC)}$$

- Here $z_G(s)$ is a polynomial with the zeros of $G(s)$ as roots, and K_{cl} is a scalar constant such that $H_{cl}(0) = 1$,

$$K_{cl} = \frac{\det(-A + BK) \det(-A + LC)}{z_G(0) \gamma(0)}$$

Example

- Consider servomotor with velocity feedback

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

- Transfer function is $G(s) = 1/(s(s+1))$, no zeros.
- Control design goal: Closed-loop poles are at $s = -2 \pm 2j$ and $K_v = 10$.

Example

- To set closed-loop poles p_1, p_2 to desired values $-2 \pm 2j$, set $K = [8 \ 3]$ by Ackermann's.
- Suppose we set observer poles to $p_3 = -0.2$ and $p_4 = -0.2$.
- Usually these poles are placed to left of closed-loop poles p_1, p_2 , but here they serve a different purpose. We'll set zeros of $\gamma(s)$ to counter effect of p_3 and p_4 .

Example

- For desired observer poles p_3 and p_4 Ackermann's gives observer gain $L = [-0.6 \ 0.64]^T$.
- Now need to select two zeros z_1 and z_2 (contributed by controller) such that

$$\frac{1}{K_v} = \frac{1}{z_1} + \frac{1}{z_2} - \frac{1}{p_1} - \frac{1}{p_2} - \frac{1}{p_3} - \frac{1}{p_4}$$
$$-10.4 = \frac{1}{z_1} + \frac{1}{z_2}$$

Example

- To satisfy

$$\frac{1}{z_1} + \frac{1}{z_2} = -10.4$$

- Try $z_1 = z_2 = -1/5.2 = -0.1923$. Note how z_1, z_2 is close to p_3, p_4 so they almost cancel and we can minimise their effect on closed-loop system.

Example

- Thus $\gamma(s) = (s-z_1)(s-z_2) \cong s^2 + 0.3846s + 0.0370$.
- Remains to specify $\beta = M/N$ and N .
- The equation to solve for $\beta = [\beta_1 \ \beta_2]^T$ is

$$\gamma(s) = \det(sI - A + BK + LC - \beta K)$$

$$s^2 + 0.3846s + 0.0370 = \det \left(\begin{bmatrix} s - 0.6 - 8\beta_1 & -1 - 3\beta_1 \\ 8.64 - 8\beta_2 & s + 4 - 3\beta_2 \end{bmatrix} \right)$$

Example

- This leads to pair of equations

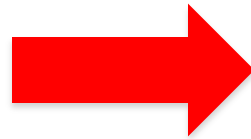
$$3.4 - 8\beta_1 - 3\beta_2 = 0.3846$$

$$6.24 - 6.08\beta_1 - 6.2\beta_2 = 0.0370$$

- The solution is

$$\beta_1 = 0.0028$$

$$\beta_2 = 0.9978$$



$$\beta = \begin{bmatrix} 0.0028 \\ 0.9978 \end{bmatrix}$$

- Remains to specify N .

Example

- Equation so $H_{cl}(0) = 1$ is (from earlier)

$$N = -\frac{1}{C(A - BK)^{-1}B(1 - K(A - LC)^{-1}(B - \beta))} = 8.6487$$

- Therefore

$$\begin{aligned} M &= \beta N \\ &= \begin{bmatrix} 0.0238 \\ 8.6294 \end{bmatrix} \end{aligned}$$

- K, L, M and N specified, design complete.